4 Pertubation theory

We now dive into time-dependent perturbation theory in non-relativistic quantum mechanics. For a more complete treatment, see for example Sakurai&Napolitano (2011). Here, I want to give you a taste of how to calculate the main observables in particle physics: cross sections and decay rates. In doing so, we will also develop a better understanding of the logic behind Feynman diagrams. Ultimately, the language of Feynman diagrams is just a pictorial representation of perturbation theory in quantum field theory, although you often see them used in more creative ways to visualize particle interactions and a variety of phenomena.

Let us start with the a generic form for Schrödinger's equation[8]:

$$i\frac{d}{dt}|\psi(t)\rangle = \hat{H}|\psi(t)\rangle\,, \tag{4.1}$$

where $\hat{H}$ is the Hamiltonian operator, which can be split into a free part $\hat{H}_0$ and an interaction part $\hat{H}_{\text{int}}$:

$$\hat{H} = \hat{H}_0 + g\hat{H}_{\text{int}} = -\frac{\nabla^2}{2m} + g\hat{H}_{\text{int}}. \tag{4.2}$$

The free part $\hat{H}_0$ describes the evolution of the system in the absence of interactions, while the interaction part $\hat{H}_{\text{int}}$ describes the effects of some interaction potential, for example. Note that I have pulled out a (small) coupling constant g in front of the interaction Hamiltonian. As we will see, the goal will be to expand the solution to Schrödinger's equation in powers of g.

In the absence of the interaction, $g = 0$, the time evolution of some state $|\psi(t)\rangle$ is simply given by the time evolution operator $e^{-i\hat{H}_0 t}$ acting on the initial state $|\psi(t=0)\rangle$:

$$|\psi(t)\rangle = e^{-i\hat{H}_0 t}|\psi(t=0)\rangle\,. \tag{4.3}$$

Let us decompose of state $|\psi\rangle$ into the eigenstates of the free Hamiltonian $\hat{H}_0$, namely the energy eigenstates labeled as $|n\rangle$ such that $\hat{H}_0|n\rangle = E_n|n\rangle$. Their eigenfunctions $\phi_n(\vec{x}) = \langle\vec{x}|n\rangle$ form a complete orthonormal basis of the Hilbert space, such that $\langle m|n\rangle = \delta_{mn}$ and $\sum_n |n\rangle\langle n| = \mathbb{1}$. We have

$$|\psi\rangle = \sum_n a_n|n\rangle\,, \text{ with } \sum_n |a_n|^2 = 1, \tag{4.4}$$

for some set of complex coefficients a_n that normalize the state $|\psi\rangle$. By the orthonormality of our basis states, we have

$$\langle n|\psi\rangle = \sum_m a_m\langle n|m\rangle = \sum_m a_m\delta_{nm} = a_n. \tag{4.5}$$

[8] Recall that the ket states $|\psi\rangle$ can give us the wavefunction by $\psi(\vec{x}) = \langle\vec{x}|\psi\rangle$. To ensure probability conservation, the Hamiltonian operator $\hat{H}$ must be Hermitian, $\hat{H} = \hat{H}^\dagger$, and the state must be normalized, $\langle\psi|\psi\rangle = 1$.

Acting the time evolution operator on $|\psi\rangle$ is a much simpler task now. The operator $e^{-i\hat{H}_0 t}$ gets distributed through the sum, goes past the complex coefficients, and acts on the eigenstates $|n\rangle$, yielding

$$\begin{aligned}|\psi(t)\rangle &= \sum_n a_n e^{-i\hat{H}t} |n\rangle \\ &= \sum_n a_n \left(\mathbb{1} - i\hat{H}t + \dots\right) |n\rangle\,, \\ &= \sum_n a_n \left(1 - iE_n t + \dots\right) |n\rangle\,, \\ &= \sum_n a_n e^{-iE_n t} |n\rangle\,,\end{aligned} \tag{4.6}$$

where we have used the fact that the eigenstates $|n\rangle$ are time-independent.

What is the probability that the state $|\psi(t)\rangle$ is unchanged after some time t? This is given by the square of the correlation function

$$P_\psi(t) = |C_\psi(t)|^2 = |\langle\psi|\psi(t)\rangle|^2 = \left|\sum_{m,n} a_m^* a_n e^{-iE_n t}\langle m|n\rangle\right|^2 = \left|\sum_n |a_n|^2 e^{-iE_n t}\right|^2 . \tag{4.7}$$

If at time $t = 0$, the state $|\psi\rangle$ is an eigenstate of the free Hamiltonian, say $|n_0\rangle$, then $a_n = \delta_{nn_0}$ and $P_\psi(t) = 1$ for all times: it does not change! However, if $|\psi\rangle$ is a superposition of different eigenstates, then the correlation function will oscillate in time, and the probability $P_\psi(t)$ will vary between 0 and 1.

In the general case where $|\psi\rangle$ is a superposition of multiple eigenstates, what is the long-time ($t \to \infty$) average of $P_\psi(t)$? Consider what happens to the cross terms after your square the sum in Eq. (4.7)

Take the particular case of a two-state system, $|\psi\rangle = a_1|1\rangle + a_2|2\rangle$. What is $P_\psi(t)$? Since $|a_1|^2 + |a_2|^2 = 1$, you can parameterize the coefficients as $a_1 = \cos\theta$ and $a_2 = \sin\theta$. Express your answer using only sine functions and use the fact that $\sin^2\theta/2 = (1-\cos\theta)/2$.

These exercises are useful to understand why we observe oscillation phenomena in quantum mechanics. As you have proven above, the fact that neutrinos oscillate, which we will study later, means that they are not produced as energy eigenstates of the Hamiltonian.

Now let us consider what happens when $g \neq 0$. We will not prove this here, but one can show that the time-evolved solutions to Schrödinger's equation can be written as

$$|\psi(t)\rangle = \sum_n a_n(t) e^{-iE_n t} |n\rangle\,. \tag{4.8}$$

The only change with respect to Eq. (4.6) is that the coefficients a_n are now time-dependent. This is not surprising, the interaction Hamiltonian can cause transitions between different eigenstates of the free Hamiltonian, and so the time evolution is a little more involved. However, the equation above still does not tell us how to find the values of $a_n(t)$, although note that it is still true that $\langle n|\psi(t)\rangle = a_n(t)$.

Let us plug Eq. (4.8) into Schrödinger's equation, Eq. (4.1):

$$i\frac{d}{dt}\sum_k a_k(t)e^{-iE_kt}|k\rangle = (\hat{H}_0 + g\hat{H}_{\text{int}})\sum_k a_k(t)e^{-iE_kt}|k\rangle\,, \tag{4.9}$$

$$\sum_k \left(i\frac{da_k(t)}{dt}e^{-iE_kt} + E_k a_k(t)e^{-iE_kt}\right)|k\rangle = \sum_k a_k(t)e^{-iE_kt}(E_k + g\hat{H}_{\text{int}})|k\rangle\,, \tag{4.10}$$

$$i\sum_k \frac{da_k(t)}{dt}e^{-iE_kt}|k\rangle = g\sum_k a_k(t)e^{-iE_kt}\hat{H}_{\text{int}}|k\rangle\,. \tag{4.11}$$

Taking the inner product with $\langle n|\,e^{+iE_nt}$, we get

$$\boxed{\frac{da_n(t)}{dt} = -ig\sum_k a_k(t)e^{-i(E_k-E_n)t}\mathcal{H}_{nk}, \quad \mathcal{H}_{fi} \equiv \langle f|\,\hat{H}_{\text{int}}\,|i\rangle\,,} \tag{4.12}$$

where we defined the transition amplitude $\mathcal{H}_{fi}$ between states $|i\rangle$ and $|f\rangle$ due to the interaction Hamiltonian $\hat{H}_{\text{int}}$. Through some quite harmless operations we have transformed Schrödinger's equation into a set of coupled first-order differential equations for the coefficients $a_n(t)$. This can be quite a complicated set of equations to solve in all generality and that is where perturbation theory comes in.

First, let us intentionally fail to solve this. If you just integrate both sides, you get

$$a_n(t) = a_n(0) - ig\sum_k \int_0^t dt' a_k(t')e^{-i(E_k-E_n)t'}\mathcal{H}_{nk}. \tag{4.13}$$

This is not a solution, as the right-hand side still depends on $a_n(t')$. Insert the expression for $a_n(t)$ back into the right-hand side, and you get an infinite series of nested integrals. But! Not all is lost, as for each insertion, you get an extra factor of the small number g. This is the key to perturbation theory: if g is small, then each successive term in this series is smaller than the previous one, and so you can get a pretty good idea of the solution by truncating the series at some order in g. This is the key idea of perturbation theory.

If the coupling g is small, you can imagine expanding the solutions $a_n(t)$ in powers of g, like you would any other function:

$$a_n(t) = a_n^{(0)}(t) + g a_n^{(1)}(t) + g^2 a_n^{(2)}(t) + \ldots \tag{4.14}$$

where $a_n^{(0)}(t)$ is the solution in the absence of the interaction, and $a_n^{(1)}(t)$, $a_n^{(2)}(t)$, etc. are the first and second order corrections due to the interaction.[9]

You can convince yourself that the solution at each order can only depend on the solutions at lower orders, such that we can solve for the coefficients iteratively. Of course, we must start somewhere. But that is easy, at zeroth order, we have no interaction, and so the coefficients are time-independent and the full solution is just ψ with the coefficients a_n, which we assume are known.

Let us take the simplest possible case where the time-independent state is just $\psi = |i\rangle$, that is, $a_n = \delta_{ni}$. In that case, the first-order solution for a given coefficient $a_f^{(1)}(t)$ (with $f \neq i$) is just

$$\begin{aligned} \frac{da_f^{(1)}(t)}{dt} &= -ige^{-i(E_i-E_f)t}\mathcal{H}_{fi} \\ \Longrightarrow \; a_f^{(1)}(t) &= -ig\int_0^t dt' e^{-i(E_i-E_f)t'}\mathcal{H}_{fi}, \end{aligned} \tag{4.15}$$

where we imposed that $a_f^{(1)}(t=0) = 0$ for $f \neq i$. The integral evaluates to

$$a_f^{(1)}(t) = g\mathcal{H}_{fi}\frac{e^{-i(E_i-E_f)t} - 1}{E_i - E_f}. \tag{4.16}$$

We also evaluate the probability of finding the system in the state $|f\rangle$ at time t (to first order in pertubation theory). It is just

$$P_{i\to f}(t) = \left|a_f^{(1)}(t)\right|^2 = g^2\left|\mathcal{H}_{fi}\right|^2 \frac{\sin^2\left(\frac{(E_f-E_i)t}{2}\right)}{\frac{(E_f-E_i)^2}{4}} \quad \overset{t\Delta E \gg 1}{\longrightarrow} \quad g^2\left|\mathcal{H}_{fi}\right|^2 2\pi t\delta(E_f - E_i), \tag{4.17}$$

This is interesting. The coefficient of $|f\rangle$ after some time t is proportional to the matrix element of the interaction Hamiltonian between the initial and final states, $\langle f|\hat{H}_{\text{int}}|i\rangle$, times a delta-function that ensures energy conservation, $E_f = E_i$. It is also proportional to the the square of the coupling constant g^2, as we would expect from a first-order calculation. The amplitude is of order g, but the probability is of order g^2.

In most applications, we are interested instead in the transition rate, which is the probability per unit time of making a transition from $|i\rangle$ to $|f\rangle$ per unit time. In this first order calculation with the simplest possible initial state, this is just

$$\boxed{\Gamma_{i\to f} = \frac{P_{i\to f}(t)}{t} = 2\pi g^2\left|\mathcal{H}_{fi}\right|^2\delta(E_f - E_i).} \tag{4.18}$$

This is known the first time we encounter Fermi's Golden Rule in its simplest form: a first order transition between an initial and final state states. It is saying that this rate $\Gamma_{i\to f}$ is proportional to the square of the matrix element of the interaction Hamiltonian between

[9] This procedure is made more precise by the Dyson series, which we will not get into here, but which is the basis for the full pertubation theory of Feynman diagrams.

the initial and final states, $\mathcal{H}_{fi} = \langle f | \hat{H}_{\text{int}} | i \rangle$, with some additional factors that ensure the correct kinematics. We will refine it over the course of these lectures to get the general form used for calculating decay rates and cross sections in particle physics.

Next order corrections Now, let us use the solution in (4.16) to compute the next order corrections to the transition amplitude, $a_f^{(2)}(t)$. The equation for $a_f^{(2)}(t)$ is

$$\frac{da_f^{(2)}(t)}{dt} = -ig\left(a_i^{(1)}(t)e^{-i(E_i-E_f)t}\mathcal{H}_{fi} + \sum_{k\neq i} a_k^{(1)}(t)e^{-i(E_k-E_f)t}\mathcal{H}_{fk} +\right), \tag{4.19}$$

where we have separated the $k = i$ term from the sum as the equation we derived for the first order coefficients assumed $f \neq i$. Plugging in the first-order solution for $a_k^{(1)}(t)$ with $a_i^{(0)}(t) = 1$, we get

$$\begin{aligned}\frac{da_f^{(2)}(t)}{dt} &= -ige^{-i(E_i-E_f)t}\mathcal{H}_{fi} - g^2\sum_k\left(\mathcal{H}_{fk}\mathcal{H}_{ki}e^{-i(E_k-E_f)t}\frac{e^{-i(E_i-E_k)t}-1}{E_i-E_k}\right). \qquad (4.20)\\ &= -ige^{-i(E_i-E_f)t}\mathcal{H}_{fi} - g^2\sum_k\left(\mathcal{H}_{fk}\mathcal{H}_{ki}\frac{e^{-i(E_i-E_f)t} - e^{-i(E_k-E_f)}}{E_i-E_k}\right).\end{aligned}$$

We rearranged the exponentials in the second term to show that that phase difference will oscillate in time, and for many states $|k\rangle$, it will average out to zero over long times. Integrating both sides, taking, again, $a_f^{(2)}(t = 0) = 0$, we can write the result in a very suggestive way:

$$a_f^{(2)}(t) = i\mathcal{M}_{fi}F(E_i, E_f, t) \qquad \Longrightarrow \qquad P_{i\to f}(t) = |a_f^{(2)}(t)|^2 = |\mathcal{M}_{fi}|^2|F(E_i, E_f, t)|^2, \tag{4.21}$$

where we defined the second-order transition amplitude

$$\mathcal{M}_{fi} = g\mathcal{H}_{fi} + g^2\sum_k\frac{\mathcal{H}_{fk}\mathcal{H}_{ki}}{E_k-E_i}, \tag{4.22}$$

and the kinematic function $F(E_i, E_f, t)$ can be evaluated in the long-time limit, $t \to \infty$, to give

$$|F(E_i, E_f, t)|^2 = 2\pi t\delta(E_f - E_i). \tag{4.23}$$

The transition amplitude $\mathcal{M}_{fi}$ is the quantity that will appear in all our calculations of decay rates and cross sections in particle physics. It is the sum of all possible ways to go from the initial state $|i\rangle$ to the final state $|f\rangle$, either directly (the first term) or through some intermediate states $|k\rangle$ (the second term). The kinematic function $F(E_i, E_f, t)$ encodes the time dependence and the energy differences between the states.

Putting everything together, at any order in perturbation theory, one can show that

$$\boxed{\Gamma_{i\to f} = 2\pi|\mathcal{M}_{fi}|^2\delta(E_f - E_i), \quad \mathcal{M}_{fi} = g\mathcal{H}_{fi} + g^2\sum_k\frac{\mathcal{H}_{fk}\mathcal{H}_{ki}}{E_k-E_i} + \dots} \tag{4.24}$$

This is now an even more general form of Fermi's Golden Rule for arbitrary order in perturbation theory. The transition amplitude $\mathcal{M}_{fi}$ is the sum of all possible ways to go from the initial state $|i\rangle$ to the final state $|f\rangle$, either directly or through intermediate states $|k\rangle$. The delta-function ensures energy conservation, $E_f = E_i$.

To illustrate this, we can draw the diagrams below to represent the different terms in the transition amplitude $\mathcal{M}_{fi}$.

$$i \text{ ——}\times\text{—— } f \qquad\qquad i \text{ ——}\times\overset{k}{\text{——}}\times\text{—— } f$$

(Diagrams representing the different terms in the transition amplitude $\mathcal{M}_{fi}$: (a) the first-order term $g\mathcal{H}_{fi}$ and (b) the second-order term $g^2 \sum_k \frac{\mathcal{H}_{fk}\mathcal{H}_{ki}}{E_k - E_i}$. The crosses represent the interaction Hamiltonian insertions, and the line between them represents the propagation through the intermediate state $|k\rangle$.)

Feynman diagrams are not too different from what you see above. They provide a pictorial representation of the terms in the transition amplitude $\mathcal{M}_{fi}$, making it easier to visualize the interactions between particles. Note the intermediate state $|k\rangle$ in the second diagram above: it comes with what we can call the propagator:

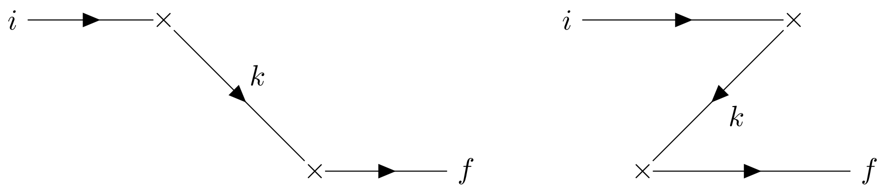
$$\text{Propagator} = \frac{1}{E_k - E_i}. \tag{4.25}$$

You may ask: should the energy not be conserved at each vertex? That is, should we not have $E_k = E_i$? The answer in non-relativistic quantum mechanics is no, as the intermediate state $|k\rangle$ is not an observable state: it is a virtual state that can have any energy. At each vertex, *momentum* is conserved, but not energy. Note that in the case of degenerate states, it may just be that $E_k = E_i$. In perturbation theory, this can be dealt with by an appropriate change of basis, but we will not get into that here.

Old Fashioned Pertubation Theory At this point I can show you something quite magical in quantum field theory. We can recover covariance by considering virtual states that go forwards, as well as *backwards* in time. Recall that through the Feynman-Stückelberg interpretation, we can interpret a particle going backwards in time as an antiparticle going forwards in time[10], so this procedure is asking us to include not just particle solutions, but also antiparticle solutions in our diagrams. Doing so by hand in quantum mechanics is called often referred to as "old-fashioned perturbation theory", but in quantum field theory, it will turn out that this procedure will be mandatory to maintain covariance and causality(!).

To see how this comes about, let me redraw the second-order diagram above in a more suggesting way, adding arrows to indicate the flow of charge with respect to time (t increases always from left to right). I will then also add by hand a second diagram where the intermediate state goes backwards in time.

[10]If this bothers you, recall the Dirac sea interpretation of antimatter as a sea of negative-energy states. Now, a plane wave with positive energy, e^{iEt}, going backwards in time, $t \to -t$, becomes a plane wave with negative energy, $e^{i(-E)t}$, going forwards in time.



(A second-order diagram with an intermediate state going forwards (left) and backwards (right) in time, the latter representing an antiparticle going forwards in time.)

Just as in the standard diagram, the propagator takes the form $1/(E_{\text{virtual}} - E_i)$. In the left-hand side, $E_{\text{virtual}} = E_k$ is the energy of the intermediate particle, but on the right-hand side, E_{virtual} is the energy of all three intermediate particles, namely: $E_{\text{virtual}} = E_i + E_k + E_f$. Since we will eventually get hit by a delta-function enforcing $E_f = E_i$, we can rewrite the propagator as

$$\text{Propagator 2} = \frac{1}{E_{\text{virtual}} - E_i} = \frac{1}{E_i + E_k + E_f - E_i} \approx \frac{1}{E_i + E_k}. \tag{4.26}$$

The amplitude for both diagrams is actually identical up to the propagator, so we can just add them up to get

$$\mathcal{M}^{(2)}_{fi} = g^2 \sum_k \mathcal{H}_{fk}\mathcal{H}_{ki} \left(\frac{1}{E_k - E_i} + \frac{1}{E_i + E_k} \right) = g^2 \sum_k \mathcal{H}_{fk}\mathcal{H}_{ki} \frac{2E_k}{E_k^2 - E_i^2}, \tag{4.27}$$

where we only write the second-order term for clarity. Up to the factor of $2E_k$ in the numerator (it will eventually find its way into the normalization of our wavefunctions to make those Lorentz invariant as well), you can show that this is a covariant propagator, as it depends on $E_k^2 - E_i^2 = p_k^2 - p_i^2$, which is explicitly Lorentz invariant as it only the depends on the square of four-momenta. Note that we had to assume momentum conservation at each vertex to get this result, $p_i^2 = E_i^2 - |\vec{p}|^2$ and $p_k^2 = E_k^2 - |\vec{p}|^2$.

Transitions to a continuum of states In the examples above, we considered transitions between discrete states $|i\rangle$ and $|f\rangle$. However, in many physical situations, the final states form a continuum, such as in scattering processes or decays into multiple particles that can be produced with a continuum distribution of momenta and energies (think of $\mu^+ \to e^+\nu_e\bar{\nu}_\mu$, for example). In such cases, we need to sum over all possible final states. To do so, we introduce the density of states $\rho(E_f)$, which counts the number of states per unit energy interval around E_f. The transition rate to a continuum of final states is then given by

$$\Gamma_{i\to\text{all } f} = \int dE_f \frac{d\Gamma_{i\to f}}{dE_f} = \int dE_f\, \rho(E_f)|\mathcal{M}_{fi}|^2 2\pi\delta(E_f - E_i). \tag{4.28}$$

Note that we replaced the transition in Eq. (4.24) by a differential transition rate $d\Gamma_{i\to f}/dE_f$ to account for the continuum of states.

What is $\rho(E_f)$? It depends on the specific system you are considering, but there are some general parameterizations we can use to count the number of states in a given energy interval. In general, the density of states can also depend on other quantities other than energy, such as the three-momentum $\vec{p}$, spin, etc. We will keep these implicit for now, but note that inside $\rho(E_f)$ you may have additional differentials to integrate over.

Let us consider a simple example of a particle in a three-dimensional box of volume $V = L^3$. As we know, the wavefunctions of the particle must satisfy boundary conditions at the walls of the box, which quantizes the allowed momenta of the particle. We can then label our solutions by integers n_x, n_y, and n_z such that the wavefunction is given by

$$\psi_{n_x,n_y,n_z}(\vec{x}) = \frac{1}{\sqrt{V}} e^{i\vec{p}_{n_x,n_y,n_z}\cdot\vec{x}}, \tag{4.29}$$

where $\vec{p}_{n_x,n_y,n_z} = \frac{2\pi}{L}(n_x, n_y, n_z)$ is the momentum of the state. Note that we have normalized the wavefunction to ensure that $\int_V d^3x|\psi|^2 = 1$. The allowed momenta are then quantized in units of $2\pi/L$, and the number of states in a momentum interval d^3p is given by

$$d^3n = dn_x dn_y dn_z = \frac{L^3}{(2\pi)^3} d^3p = \frac{V}{(2\pi)^3} d^3p. \tag{4.30}$$

The density of states in our energy interval dE_f can then be parameterized as,

$$\rho(E)dE = d^3n = \frac{V}{(2\pi)^3} d^3p = \frac{V}{(2\pi)^3} |\vec{p}|^2 d|\vec{p}| d\Omega = \frac{V}{(2\pi)^3} \frac{|\vec{p}|^2}{\beta} dE d\Omega, \tag{4.31}$$

where we also got the density of states in terms of the angle of the outgoing state f as well thorugh $d\Omega = d\cos\theta d\phi$.

Inserting this into the expression for the transition rate, recovering the f index, and calculating the cross section ($\sigma = \Gamma/\phi$ with $\phi = \beta_i/V$ the flux of incoming particles), we get

$$\sigma_{i\to \text{all } f} = \int dE_f d\Omega_f \frac{V^2}{(2\pi)^2} \frac{|\vec{p}_f|^2}{\beta_i \beta_f} |\mathcal{M}_{fi}|^2 \delta(E_f - E_i) = \int d\Omega_f \frac{V^2}{(2\pi)^2} E_i^2 |\mathcal{M}_{fi}|^2, \tag{4.32}$$

where it is now understood that the amplitude is evaluated at $E_f = E_i$ due to the delta-function and we took $E_i = |\vec{p}_i/\beta_i|$ and enforced $\beta_i = \beta_f$. One may also be interested in the differential cross section $d\sigma/d\Omega_f$,

$$\frac{d\sigma_{i\to \text{all } f}}{d\Omega_f} = \frac{V^2}{(2\pi)^2} E_i^2 |\mathcal{M}_{fi}|^2. \tag{4.33}$$

To be clear: this cross section is for the scattering of some initial state $|i\rangle$ on a finite region of some potential described by $H_{\text{int}}(\vec{x})$ to all possible final states in the continuum and multiple final state configurations $|f\rangle$, which now just depend on the angle of the outgoing state with respect to the z-direction. It requires a leap of faith only in the way that I dealt with the incoming and outgoing states in a finite volume V, but as we are about to see, these factors of volume will eventually drop out once we evaluate the amplitude $\mathcal{M}_{fi}$ in terms of properly normalized wavefunctions.

Let us investigate the amplitude a little more. Consider the incoming and the outgoing states to be single-particle states with momenta $\vec{p}_i$ and $\vec{p}_f$, respectively. The properly normalized wavefunctions for these states at some time t are given by

$$\psi_i(\vec{x}) = \frac{1}{\sqrt{V}} e^{-i(E_i t - \vec{p}_i \cdot \vec{x})}, \quad \psi_f(\vec{x}) = \frac{1}{\sqrt{V}} e^{-i(E_f t - \vec{p}_f \cdot \vec{x})}. \tag{4.34}$$

The matrix element of the interaction Hamiltonian between these states is then given by

$$\begin{aligned} \mathcal{M}_{fi} = g\mathcal{H}_{fi} = g \langle f | \hat{H}_{\text{int}} | i \rangle &= g \int_V d^3x\, \psi_f^*(\vec{x}) \hat{H}_{\text{int}} \psi_i(\vec{x}) \\ &= g \frac{e^{-i(E_i - E_f)t}}{V} \int_V d^3x\, \hat{H}_{\text{int}} e^{-i\vec{q}\cdot\vec{x}} = g \frac{e^{-i(E_f - E_i)t}}{V} \widetilde{H}_{\text{int}}(\vec{q}), \end{aligned} \tag{4.35}$$

with $\vec{q} = \vec{p}_f - \vec{p}_i$ the momentum transfer and $\widetilde{H}_{\text{int}}(\vec{q}) = \int_V d^3x\, e^{-i\vec{q}\cdot\vec{x}} \hat{H}_{\text{int}}(\vec{x})$ the Fourier transform of the interaction Hamiltonian.

This is highly non-trivial: the amplitude factor in the transition rate is proportional to the Fourier transform of the interaction Hamiltonian as a function of the momentum transfer $\vec{q}$. This is quite a general result any wavefunction can be decomposed into plane waves, and so the interaction Hamiltonian shows up in the rates in Fourier space.

Putting it all together, we get the differential cross section:

$$\boxed{\frac{d\sigma_{i\to\text{all } f}}{d\Omega} = \frac{g^2 E_i^2}{(2\pi)^2} |\widetilde{H}_{\text{int}}(\vec{q})|^2.} \tag{4.36}$$

What we have done so far contains the main ingredients to compute scattering cross sections in particle physics, although it is still done within the framework of a non-relativistic theory. If you understand this derivation, you will have quite a lot under your belt to tackle the relativistic case, which you can easily pick up from standard textbooks.

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5 Potential scattering

What kinds of interaction Hamiltonians $\hat{H}_{\text{int}}(\vec{x})$ can we consider? The simplest case is a static potential, $\hat{H}_{\text{int}}(\vec{x}) = V(\vec{x})$ (we already implicitly assumed that was the case above). For example, the Coulomb potential between two charged particles is given by

$$V_{\text{Coulomb}}(r) = \alpha \frac{q_1 q_2}{r}, \tag{5.1}$$

where $\alpha \simeq 1/137$ is the fine-structure constant and $r = |\vec{x}|$ is the distance between the two particles. In our scattering problem above, you can imagine a very heavy nucleus of charge Z at the origin, which generates a Coulomb potential $V(r) = \alpha Z/r$, and a light charged particle (like an electron or a proton) scattering off this potential. At infinitely early times, the light particle is far away from the nucleus and does not feel its presence, but as it gets closer, it feels the Coulomb potential and gets deflected. The differential cross section will then give you the probability of the light particle being scattered into a given angle.

A massive particle mediator, such as the W and Z bosons of the weak interaction, gives rise to a Yukawa potential:

$$V_{\text{Yukawa}}(r) = \alpha_Y \frac{q_1 q_2}{r} e^{-m_{\text{med}} r}, \tag{5.2}$$

where m_{med} is the mass of the mediator particle. The exponential factor $e^{-m_{\text{med}} r}$ causes the potential to fall off exponentially for $m_{\text{med}} r > 1$, much faster than the $1/r$ of the Coulomb potential, making the interaction have an effective range of $r \sim 1/m_{\text{med}}$. In this case, you can imagine a neutrino scattering off a nucleon through the exchange of a W or Z boson. Only when the neutrino is very close to the nucleon, at distances $r \lesssim 1/m_{W,Z}$, does it feel the potential. We can then ask the question: what is the probability of the neutrino being scattered into a given angle after interacting with the nucleon?

> One way to see where the Yukawa potential comes from is to solve the wave equation for a scalar field $\phi(\vec{x}, t)$ with mass m_{med}. This scalar will play the role of the mediator particle, although for this class, you will just have to believe me that such a field description is indeed appropriate to describe the potential felt by a test particle. The equation for this mediator scalar field is called the *Klein-Gordon* equation:
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> $$\partial_\mu \partial^\mu \phi(\vec{x}, t) + m_{\text{med}}^2 \phi(\vec{x}, t) = 0 \quad \Rightarrow \quad (\partial_t^2 - \nabla^2 + m_{\text{med}}^2)\phi(t, \vec{x}) = 0. \tag{5.3}$$
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> For simplicity, we consider a static solution, $\partial_t \phi(\vec{x}, t) = 0$, and assume spherical symmetry, $\phi(t, \vec{x}) = \phi(r = |\vec{x}|)$. By a change of variables to spherical coordinates and two trivial angular integrations, we get
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> $$-\frac{1}{r}\partial_r^2 (r\phi(r)) + m_{\text{med}}^2 \phi(r) = 0. \tag{5.4}$$
>
> A solution to this equation can be found by inspection:
>
> $$\phi(r) = \frac{Ae^{-m_{\text{med}} r}}{r} + \frac{Be^{m_{\text{med}} r}}{r}, \tag{5.5}$$
>
> where A and B are constants determined by boundary conditions. The second term diverges as $r \to \infty$, so we set $B = 0$. The first term is the Yukawa potential, which falls off exponentially with distance. The massless limit, $m_{\text{med}} \to 0$, recovers the Coulomb potential.

Scattering off a static Yukawa potential As a simple example, let us consider the case where the interaction Hamiltonian is just the Yukawa potential, $\hat{H}_{\text{int}}(\vec{x}) = V_{\text{Yukawa}}(r)$. (Note that the coupling constant of the interaction that we expanded on in pertubation theory is just α_Y, so it is already inside the potential.) The Fourier transform of this potential is given by

$$\widetilde{V}_{\text{Yukawa}}(\vec{q}) = \int d^3x\, e^{-i\vec{q}\cdot\vec{x}} V_{\text{Yukawa}}(r) = \alpha_Y q_1 q_2 \int d^3x\, \frac{e^{-i\vec{q}\cdot\vec{x}} e^{-m_{\text{med}} r}}{r} \tag{5.6}$$
$$= 2\pi\alpha_Y q_1 q_2 \int_0^\infty dr\, r e^{-m_{\text{med}} r} \int_{-1}^{1} d\cos\theta\, e^{-i|\vec{q}| r\cos\theta} = \frac{4\pi\alpha_Y q_1 q_2}{|\vec{q}|^2 + m_{\text{med}}^2},$$

where for this integration we aligned the z-axis with the momentum transfer $\vec{q}$ to simplify things. Using

$$|\vec{q}|^2 = |\vec{p_f} - \vec{p_i}|^2 = |p_f|^2 + |p_i|^2 - 2|p_f||p_i|\cos\theta = 2|\vec{p}|^2(1-\cos\theta) = 4\beta_i^2 E_i^2 \sin^2(\theta/2), \tag{5.7}$$

the differential cross section for scattering off a static Yukawa potential is then given by

$$\boxed{\frac{d\sigma}{d\Omega} = \frac{E_i^2}{(2\pi)^2}\left|\widetilde{V}_{\text{Yukawa}}(\vec{q})\right|^2 = \frac{\alpha_Y^2 q_1^2 q_2^2}{4E_i^2} \frac{1}{\left(\beta_i^2 \sin^2(\theta/2) + \frac{m_{\text{med}}^2}{4E_i^2}\right)^2}}. \tag{5.8}$$

In the massless limit, $m_{\text{med}} \to 0$, and taking $m_i = m_f = 0$, we recover the Rutherford scattering cross section for a Coulomb potential:

$$\frac{d\sigma}{d\Omega} = \frac{\alpha^2 Z^2}{4E_i^2 \sin^4(\theta/2)}. \tag{5.9}$$

Note that the cross section diverges as $|\vec{q}| \to 0$, which corresponds to forward scattering ($\theta \to 0$). This divergence is a well-known feature of Coulomb scattering and is related to the long-range nature of the Coulomb potential. In reality, other effects, such as screening by other charges or the finite size of the scattering particles, will regularize this divergence.

Now, consider the opposite case where the mediator is very heavy, $m_{\text{med}} \gg |\vec{q}|$ and take $q_1 = q_2 = 1$. In this case, the cross section simplifies to

$$\frac{d\sigma}{d\Omega} \approx \alpha_Y^2 \frac{4E_i^2}{m_{\text{med}}^4}. \tag{5.10}$$

The cross section grows with energy(!). This is a common feature of weak interactions mediated by heavy particles, such as the W and Z bosons in the Standard Model. Note that this growth cannot continue indefinitely, as eventually, the largest values of $|\vec{q}|$ will dominate the denominator, and the cross section will plateau.

> Electromagnetic processes prefer small momentum transfers, so they dominate at low energies. Weak processes, on the other hand, prefer large momentum transfers, so they increase in importance at high energies. The Weak force is weak because we usually probe it at low energies, where the W and Z bosons are very heavy compared to the momentum transfers involved. They tend to suppress decay rates and cross sections through their large masses in the propagators.